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Key Points:

- Drought propagation mechanisms and their controlling factors are analyzed in different regions of the CONUS
- The spatial pattern of hydrologic drought characteristics is found to coincide with the pattern of climate properties
- The storage-discharge relationship is identified as a key watershed property affecting drought propagation

Supporting Information:

- Supporting Information S1

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Drought Propagation in Contiguous U.S. Watersheds: A Process-Based Understanding of the Role of Climate and Watershed Properties

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Abstract Drought impacts on the society and environment are influenced by how droughts propagate through the hydrologic cycle; however, the physical processes involved in drought propagation are not well understood. In this study, a physically based hydrologic model is used to understand the drought propagation mechanisms and their controlling factors in multiple watersheds selected from different regions of the contiguous United States (CONUS). The characteristics of hydrologic droughts are found to be significantly different in three regions of the CONUS: (1) western U.S. watersheds have long and intense streamflow droughts, (2) Great Plains and southwest U.S. watersheds have low intensity and duration of streamflow droughts, and (3) eastern U.S. watersheds have short but intense streamflow droughts. This spatial pattern of hydrologic drought characteristics is found to coincide with the pattern of climate properties. Highly seasonal recharge driven by winter precipitation or snowmelt leads to longer streamflow droughts in the western United States; whereas humid climate with low seasonality in the eastern United States leads to shorter streamflow droughts. Furthermore, the storage-discharge relationship is identified as a key watershed property which controls the intensity of hydrologic droughts. Higher sensitivity of baseflow to watershed storage in the western and the eastern U.S. watersheds leads to more intense streamflow droughts in these regions. In the Great Plains and southwest United States, watersheds have a lower sensitivity of baseflow to catchment storage which, combined with the highly arid climate, leads to low variability in baseflow, and in turn results in low intensity of streamflow droughts in the region.

1. Introduction

Droughts have been recognized as one of the most complex natural disasters, especially because they involve interactions between numerous climatological and hydrological processes, which gradually evolve with time (Wilhite & Glantz, 1985). Droughts can be classified into meteorological, agricultural (soil moisture), and hydrologic droughts depending on the components of the hydrological cycle which are affected by droughts (Wilhite & Glantz, 1985). Meteorological droughts are defined as periods with deficient precipitation; deficits in soil moisture and streamflow are referred to as agricultural and hydrologic droughts, respectively. The process of translation of droughts through different components of the hydrologic cycle is known as drought propagation (Eltahir & Yeh, 1999; Wang et al., 2011). The processes involved in drought propagation ultimately determine the characteristics of water shortages experienced at a given place during a drought event, and these processes are influenced by both climate and watershed properties (Van Loon & Van Lanen, 2012). The purpose of this study is to improve the understanding of drought propagation and its influencing factors, which will help better characterize the impacts of droughts on water resources.

A number of studies have attempted to understand how the drought propagation process is influenced by climate and watershed properties. These studies can be categorized into two groups, i.e., those using statistical analysis and those using hydrologic modeling. Studies based on statistical analysis have used various statistical techniques such as correlation analysis (Barker et al., 2016; Haslinger et al., 2014; Huang et al., 2017; Jiang et al., 2015; Liu et al., 2019; Peña-Gallardo et al., 2019; Van Loon & Laaha, 2015; Wang et al., 2020; Wu et al., 2018; Xu et al., 2019; Yang et al., 2017) or machine learning techniques (Konapala & Mishra, 2020; Veetil & Mishra, 2020; Veetil et al., 2018) for identifying the climate and watershed properties which control drought propagation. The main advantage of the statistical approach is that it allows researchers to identify the dominant controlling factors from a large number of watershed and climate

properties. Studies have shown that the most important watershed property controlling drought propagation varies from place to place, e.g., watershed elevation in the watersheds of United States (Konapala & Mishra, 2020; Peña-Gallardo et al., 2019), land-use properties in loess plateau region of China (Wu et al., 2018), and geological properties of watersheds in Austria (Haslinger et al., 2014). While statistical techniques can reveal which factors have the most significant influence on drought propagation in a given region, the drawback of this approach, as being well noted, is that it is difficult to understand the physical mechanisms, i.e., how the identified controlling variables influence drought propagation, which makes it difficult to generalize the findings of these studies (Marsh et al., 2007; Van Loon & Van Lanen, 2012).

Modeling-based studies utilize hydrologic modeling for understanding the physical mechanisms of drought propagation under different climate and watershed properties (Apurv et al., 2017; Gevaert et al., 2018; Longobardi & Van Loon, 2018; Melsen & Guse, 2019; Staudinger et al., 2015; Tallaksen et al., 2009; Van Lanen et al., 2013; Van Loon et al., 2014; Van Loon & Van Lanen, 2012). The main drawbacks of the modeling approach lie in the limitations of hydrologic models in representing the complexities of real-world watersheds and the uncertainties in model parameters and outputs. Nevertheless, hydrologic models have been successfully applied for conceptual understanding of drought propagation mechanisms and for the development of classification frameworks for hydrologic droughts based on climate and watershed properties. For example, Van Loon and Van Lanen (2012) used hydrologic modeling to develop a process-based classification of hydrologic droughts. Apurv et al. (2017) and Van Loon et al. (2014) have used modeling experiments with “virtual watersheds”, in which drought propagation was studied by systematically varying climate properties to understand how climate properties like aridity, seasonality, and timing of rainfall influence drought propagation under constant hydrologic model parameters. Similarly, Van Lanen et al. (2013) studied the influence of soil and groundwater properties on drought propagation by changing soil and groundwater parameters for a fixed climate type in a conceptual hydrologic model.

Previous studies on drought propagation in the contiguous United States (CONUS) have been conducted using statistical approaches (e.g., Konapala & Mishra, 2020; Peña-Gallardo et al., 2019; Veetil et al., 2018). These studies have focused on identifying the climate and watershed properties that control drought propagation in the CONUS watersheds; however, the patterns characterizing the drought propagation mechanisms across the CONUS, the differences, and their controls among different regions are not well understood. In this study, we aim to fill this knowledge gap by using a physically based hydrologic model to analyze the hydrologic processes involved in drought propagation in watersheds selected from different regions of the CONUS. We first identify the patterns characterizing the drought propagation mechanisms across the CONUS and the differences among different regions; following which we seek understanding on how the climate and watershed properties give rise to these different mechanisms. Such understanding will be informative for developing drought management strategies suitable to different climate and watershed properties.

The remainder of the paper is organized as follows. Section 2 describes the data sets used in the study and the methodology adopted for studying drought propagation. Section 3 presents the results of the study including the spatial patterns of hydrologic drought characteristics and the key differences in hydrologic processes during drought events in different regions of the CONUS. Section 4 discusses the results of the study in the context of findings of previous drought propagation studies as well as the limitations of this study. Section 5 concludes with a summary of the main findings.

2. Materials and Methods

2.1. Data

In this study, we use the Livneh et al. (2013) data set which is derived from the simulation of the Variable Infiltration Capacity (VIC) land surface hydrology model at a 3-hr time-step for the CONUS at a 1/16° resolution for the 1915–2011 period. The VIC model (Liang et al., 1994) simulates both the terrestrial water and energy balances at the grid scale. The Livneh et al. (2013) data set has been derived by forcing the VIC model with observed data sets for precipitation and temperature and observation-derived data sets for solar radiation and wind speed for the 1915–2011 period. The soil and land cover data for the VIC model were specified using the Land Data Assimilation System (LDAS) North America Project data set (Mitchell et al., 2004). The soil characteristics are derived from the 1-km STATSGO database (Miller & White, 1998) which is based on

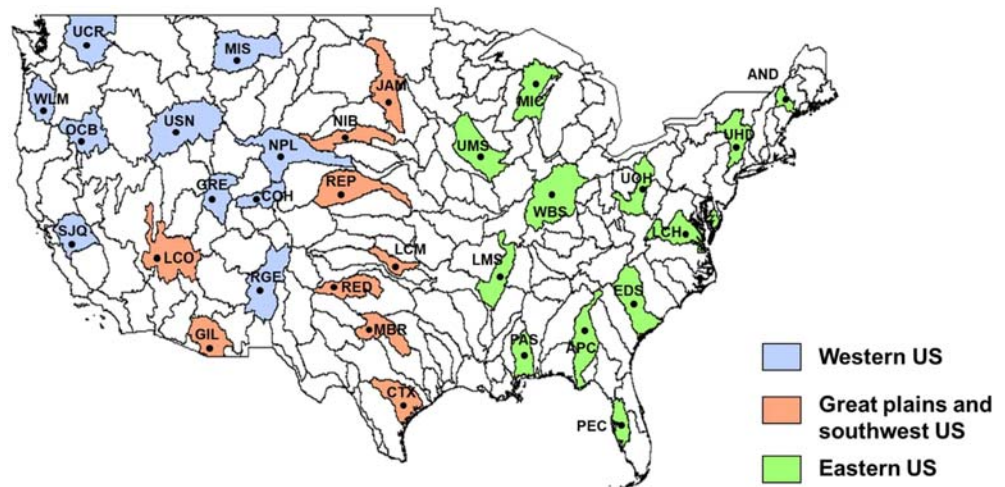


Figure 1. HUC4 watersheds considered in the study and the three regions identified with different hydrologic drought characteristics.

the works of Cosby et al. (1984), Rawls et al. (1998), and Reynolds et al. (2000). The land cover was specified using Hansen et al. (2000) data set, whereas multiple data sets were used for specifying land cover characteristics such as leaf area index (Myneni et al., 1997), rooting depth (Jackson et al., 1996), architectural resistance (Ducoudré et al., 1993), and stomatal resistance (Defries & Townshend, 1994). The comparison of simulated streamflow with the observed streamflow data has shown that the simulated outputs of the VIC model match the observed streamflow quite well for the major river basins of the United States (Livneh et al., 2013).

In the VIC model, the storage is divided into three layers: a canopy layer, a shallow layer, and a deep layer. The shallow layer contributes to surface runoff, which is modeled by incorporating subgrid variability in soil storage capacity by representing it as a spatial probability distribution. Evaporation only occurs in the top layer whereas transpiration can take place from all layers depending on the rooting depth. The shallow layer contributes to recharge of the deep layer storage, which is modeled using the Brooks and Corey (1964) relationship. The baseflow is modeled from the deep storage layer using a nonlinear recession function, based on the formulation of the Arno model (Franchini & Pacciani, 1991). The streamflow is the sum of baseflow and surface runoff. The VIC model has been successfully used for many hydrologic studies in different regions of the United States (Andreadis et al., 2005; Badger et al., 2018; Barnhart et al., 2016; Demaria et al., 2016; Sheffield et al., 2004; Xiao et al., 2018).

The data for watershed properties were accessed from the LDAS data set, which are used for specifying model parameters for the VIC model simulations in the Livneh et al. (2013) data set: soil properties were taken from <https://ldas.gsfc.nasa.gov/nldas/soils>, land cover data were taken from <https://ldas.gsfc.nasa.gov/index.php/nldas/vegetation-class>, and elevation data were taken from <https://ldas.gsfc.nasa.gov/nldas/elevation>. The Livneh et al. (2013) data set was accessed through the following websites, the precipitation data were taken from <https://psl.noaa.gov/data/gridded/data.livneh.metvars.html> and the VIC model outputs were taken from <https://psl.noaa.gov/data/gridded/data.livneh.fluxvars.html>.

2.2. Study Regions

We use the VIC model outputs to analyze the water-budget during droughts for U.S. Geological Survey (USGS) hydrologic unit code 4 (HUC4) watersheds selected from different regions of the CONUS (Figure 1). In order to make the selected watersheds to represent the varied climate properties across the CONUS, we use the ecoregion classification proposed by Bukovsky (2011), which has been shown to be a good proxy for the regional climatology in the country (Figure S1 of the supporting information). Within each climate region, we select watersheds with different elevation, soil type, and land cover to represent variations in watershed properties (Figure S1). For example, in the southeast U.S. region, we select APC watershed with loamy sand soil and woodland cover, PEC watershed with sandy soil and wooded

Table 1
Selected HUC4 Watersheds and Their Properties (Source: LDAS Data Set)

HUC4 ID	Abbreviation	Name	Area (1,000 km ²)	Mean elevation (m)	Mean slope (%)	Dominant soil type	Dominant land cover
1709	WLM	Willamette	29.7	552	0.78	Silty clay loam	Evergreen Needleleaf forest
1702	UCR	Upper Columbia	72.0	855	0.56	Silt loam	Evergreen Needleleaf forest
1712	OCB	Oregon Closed Basins	45.1	1,517	0.39	Loam	Open Shrubland
1804	SJQ	San Joaquin	40.9	840	1.37	Sandy loam	Cropland
1004	MIS	Missouri-Musselshell	60.8	1,082	0.42	Loam	Grassland
1704	USN	Upper Snake	92.9	1,857	0.71	Loam	Grassland
1401	COH	Colorado headwaters	25.4	2,567	0.89	Loam	Grassland
1018	NPL	North Platte	80.4	1,842	0.48	Sandy loam	Grassland
1406	GRE	Lower Green	37.7	2,088	1.00	Loam	Open Shrubland
1302	RGE	Rio Grande-Elephant Butte	70.2	2,088	0.67	Loam	Open Shrubland
1505	GIL	Middle Gila	46.5	1,058	0.57	Loam	Closed Shrubland
1501	LCO	Lower Colorado-Lake Mead	78.4	1,475	0.77	Sandy loam	Open Shrubland
1025	REP	Republican	64.5	973	0.19	Silt loam	Grassland
1016	JAM	James	54.7	461	0.15	Loam	Cropland
1015	NIB	Niobrara	37.1	953	0.20	Sand	Grassland
1112	RED	Red headwaters	38.6	882	0.25	Clay loam	Grassland
1105	LCM	Lower Cimarron	18.2	391	0.12	Silt loam	Grassland
1206	MBR	Middle Brazos	40.4	392	0.15	Clay loam	Wooded grassland
1210	CTX	Central Texas Coastal	47.1	148	0.16	Clay	Wooded grassland
0708	UMS	Upper Mississippi-Iowa	59.4	279	0.06	Silty clay loam	Cropland
0802	LMS	Lower Mississippi-St. Francis	41.8	85	0.05	Silt loam	Cropland
0403	MIC	Northwestern Lake Michigan	48.4	311	0.15	Sand	Deciduous Broadleaf Forest
0512	WBS	Wabash	85.3	211	0.06	Silt loam	Cropland
0503	UOH	Upper Ohio	34.6	317	0.11	Silt loam	Deciduous Broadleaf Forest
0317	PAS	Pascagoula	32.0	73	0.06	Sandy loam	Woodland
0313	APC	Apalachicola	53.4	147	0.11	Loamy sand	Woodland
0310	PEC	Peace-Tampa Bay	27.5	20	0.02	Sand	Wooded grassland
0305	EDS	Edisto-Santee	61.3	158	0.21	Sandy loam	Woodland
0208	LCH	Lower Chesapeake	53.5	200	0.24	Sandy loam	Deciduous Broadleaf Forest
0104	AND	Androskoggin	9.1	390	0.37	Sandy loam	Mixed Cover
0202	UHD	Upper Hudson	32.8	344	0.46	Silt loam	Mixed Cover

grassland cover, and LMS watershed with silty loam soil and cropland cover. In the Midwest region, we select WBS watershed with silty loam soil and cropland cover, UMS watershed with silty clay loam soil, and MIC watershed with sandy loam soil and deciduous broadleaf forest cover. Similarly, in the Great Plains region we consider JAM and MIS watersheds from the Northern Plains region with loamy soil, REP and LCM from the Central Plains with silty loam soil, and RED and MBR watersheds from the Southern Plains with clayey loam soil. There is a significant elevation gradient in the watersheds of western United States and therefore we select watersheds from varying altitudes; high elevation watersheds COH, NPL, RGE, GRE, and USN located in the Rocky mountains and UCR, WLM, SJQ, LCO, and GIL watersheds with lower elevations located in the northwest and southwest United States. The properties of the selected watersheds such as area, elevation, soil type, and land cover are shown in Table 1. These properties have been taken from the LDAS data set, which is used for specifying the parameters for VIC model simulations in the Livneh et al. (2013) data set. For the selected watersheds, we aggregate the VIC model outputs by averaging the values for all grid points within a watershed boundary; we then analyze the aggregated outputs at the monthly scale to study the process of drought propagation.

2.3. Drought Propagation Ratios

In order to identify drought events, we use the threshold-based approach in which a drought event is defined as the period when the variable of interest is below a predefined threshold (Yevjevich, 1983). Since many watersheds exhibit seasonal variations of hydro-meteorological variables, we employ a monthly varying threshold to define drought events, which is represented as percentile values of the historical distributions

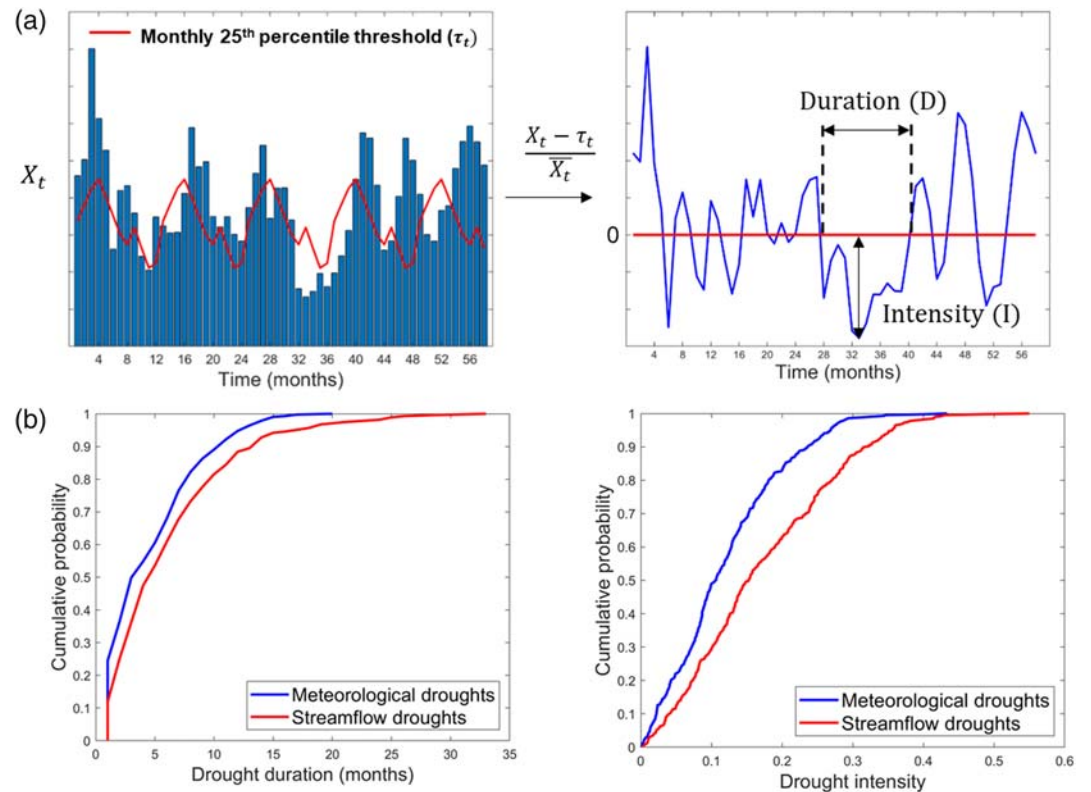


Figure 2. (a) Transformation of raw time series to a standardized time-series using a monthly 25th percentile threshold and calculation of drought duration and intensity; (b) comparison of cumulative distribution functions of durations and intensities of meteorological and hydrologic droughts.

of the variable in each month. Most previous studies have used threshold levels in the range of 5th to 30th percentile for drought studies (Fleig et al., 2006; Van Loon et al., 2014; Van Loon & Laaha, 2015; Van Loon & Van Lanen, 2012). In this study, we use the monthly 25th percentile (probability of non-exceedance 25%) as the threshold for defining droughts. We also perform sensitivity analysis of the results with respect to the threshold, which is discussed in the following section.

Following the monthly 25th percentile threshold, an example of the drought identification and calculation of drought duration and intensity is shown in Figure 2, which is based on the method proposed in Van Loon et al. (2014). We first standardize the raw time-series (X_t) by subtracting the threshold (τ_t) and dividing by the long-term mean of the variable ($\bar{X}_t$), where X_t is the hydro-meteorological variable for which droughts are to be identified (Figure 2a). Droughts are then identified as periods when the standardized time-series is less than zero. We compute two characteristics of the identified drought events: (1) drought duration (D), defined as the number of months a drought event lasts, and (2) drought intensity (I), defined as the maximum absolute deviation of the standardized time-series from zero during a drought event. We apply this procedure to the precipitation time-series to identify meteorological droughts and to the streamflow time-series to identify hydrologic droughts, and then compare the empirical cumulative distribution functions (ECDFs) of durations and intensities of meteorological and hydrologic droughts as shown in Figure 2b. The number of precipitation and streamflow drought events identified for each watershed are shown in Table S1 of the supporting information.

We use propagation ratios (Apuv et al., 2017) for comparing the CDFs of meteorological and hydrologic droughts. The propagation ratio for drought intensity (R_I) can be computed as the geometric mean of the

ratios of 10 equally spaced quantiles (deciles) of hydrologic drought intensity to that of meteorological drought intensity:

$$R_I = \left(\prod_{j=1}^{10} \frac{qI_j^{\text{streamflow}}}{qI_j^{\text{precipitation}}} \right)^{\frac{1}{10}}, \quad (1)$$

where $qI_j^{\text{streamflow}}$ is the j^{th} quantile of the CDF of streamflow drought intensity and $qI_j^{\text{precipitation}}$ is the j^{th} quantile of the CDF of precipitation drought intensity. Similarly, the propagation ratio for drought duration can be computed as

$$R_D = \left(\prod_{j=1}^{10} \frac{qD_j^{\text{streamflow}}}{qD_j^{\text{precipitation}}} \right)^{\frac{1}{10}}, \quad (2)$$

where $qD_j^{\text{streamflow}}$ is the j^{th} quantile of the CDF of streamflow drought duration and $qD_j^{\text{precipitation}}$ is the j^{th} quantile of the CDF of precipitation drought duration. The propagation ratios represent how intense or how long the hydrologic droughts are relative to meteorological droughts in a watershed. Large values of R_D or R_I indicate that hydrologic droughts have higher intensity or duration than meteorological droughts and vice versa. In this study, we also compute the propagation ratios for baseflow, similar to Equations 1 and 2, to understand the role of subsurface flow in drought propagation.

2.4. Climate and Watershed Properties

In this study, we use certain indices to characterize the climate and watershed properties in the selected watersheds. The indices for representing the climate properties are as follows:

1. Aridity Index (AI): defined as the long-term average annual potential evapotranspiration $\overline{ET_p}$ divided by the long-term average annual precipitation $\bar{P}$ (Budyko et al., 1974):

$$AI = \frac{\overline{ET_p}}{\bar{P}}. \quad (3)$$

2. Phase index (PI): represents the timing of precipitation with respect to potential ET within a year (Woods, 2009). To calculate PI, first monthly precipitation and potential ET are expressed as sine functions:

$$P(t) = \bar{P} \left[1 + \delta_P \sin \left(\frac{2\pi(t - s_P)}{\tau_P} \right) \right] \quad (4)$$

and

$$ET_p(t) = \overline{ET_p} \left[1 + \delta_{ET_p} \sin \left(\frac{2\pi(t - s_{ET_p})}{\tau_{ET_p}} \right) \right], \quad (5)$$

where t is time (months), s is a phase shift (months), τ is duration of the seasonal cycle (12 months), δ is the dimensionless amplitude, and the subscripts P and ET_p stand for precipitation and potential ET, respectively. The phase index is then calculated as

$$PI = \delta_P \delta_{ET_p} \cos \left(\frac{2\pi(s_P - s_{ET_p})}{\tau} \right). \quad (6)$$

The dimensionless index PI describes the degree to which precipitation is in phase with the potential ET regime. PI can have values between -1 (strongly winter dominant precipitation or out-of-phase climate) and $+1$ (strongly summer dominant precipitation or in-phase climate).

3. Seasonality index (SI): represents the degree of variability in monthly precipitation within a year (Walsh & Lawler, 1981), which is defined as follows:

$$SI = \frac{1}{\bar{R}} \sum_{n=1}^{12} \left| \bar{x}_n - \frac{\bar{R}}{12} \right|, \quad (7)$$

where $\bar{x}_n$ is the mean monthly precipitation for the month n and $\bar{R}$ is the mean annual precipitation. High values of SI represent highly seasonal precipitation whereas low values of SI show the lack of seasonality in precipitation.

The following three indices are used to characterize the water-balance in the selected watersheds:

1. Baseflow Index (BFI): defined as the fraction of streamflow that is contributed by baseflow, i.e., the ratio of baseflow to total streamflow. We use the VIC-simulated baseflow to calculate the BFI.
2. Discharge sensitivity to storage: We use the ratio of standard deviation of monthly VIC-simulated baseflow to the standard deviation of monthly watershed deep layer storage, as a measure of how strongly the streamflow varies in response to the variation in the watershed storage. Here the ratio of standard deviations of baseflow and storage serves as a measure of sensitivity of baseflow to storage because in the VIC model, the baseflow is modeled as a function of deep layer storage, which ensures a causal relationship between the two variables.
3. Partial correlation between snowmelt and recharge given precipitation: We compute the partial correlation between snowmelt (s) and recharge (r) after removing the effects of precipitation (p) $\rho_{sr,p}$ to quantify the contribution of snowmelt to recharge generation in the selected watersheds. Partial correlation is calculated by calculating the correlation coefficient between the residuals obtained by regressing snowmelt and recharge against precipitation, thereby removing the effects of precipitation on both the variables. In this way, partial correlation allows the quantification of the influence of individual controlling variables even if the variables are correlated (Johnson, 1978; Jukić & Denić-Jukić, 2015; Pizarro & Lall, 2002; Treble et al., 2005).

The climate and watershed properties of the selected watersheds are shown in Figure 3. As can be seen, the aridity of climate is high in the watersheds of western and central United States and low in eastern and northwestern United States (Figure 3a). While the watersheds in western United States have a winter-dominated precipitation regime, the watersheds in central United States receive most of the precipitation in the summer (Figure 3b). The seasonality of precipitation is highest in central U.S. watersheds and low in the Rocky Mountains and eastern U.S. watersheds (Figure 3c). In terms of watershed properties, the eastern U.S. watersheds have the strongest sensitivity of discharge with respect to deep-layer storage, followed by the western U.S. watersheds, whereas the watersheds in the Great Plains and southwest United States have very low sensitivity of discharge relative to the variations in the deep-layer storage (Figure 3d). The VIC-simulated BFI is highest in eastern U.S. watersheds followed by western U.S. watersheds and lowest in the Great Plains watersheds (Figure 3e). Snowmelt plays an important role in the recharge generation in the watersheds in the Rocky Mountain region and in watersheds of Great Lakes region and northeast United States (Figure 3f).

3. Results

3.1. Spatial Patterns of Drought Propagation

The propagation ratios for drought duration and intensity calculated for the selected watersheds are shown in Figure 4. The propagation ratios are calculated for both baseflow and streamflow (sum of baseflow and surface runoff) to understand the roles of surface and subsurface flows in drought propagation. A distinct

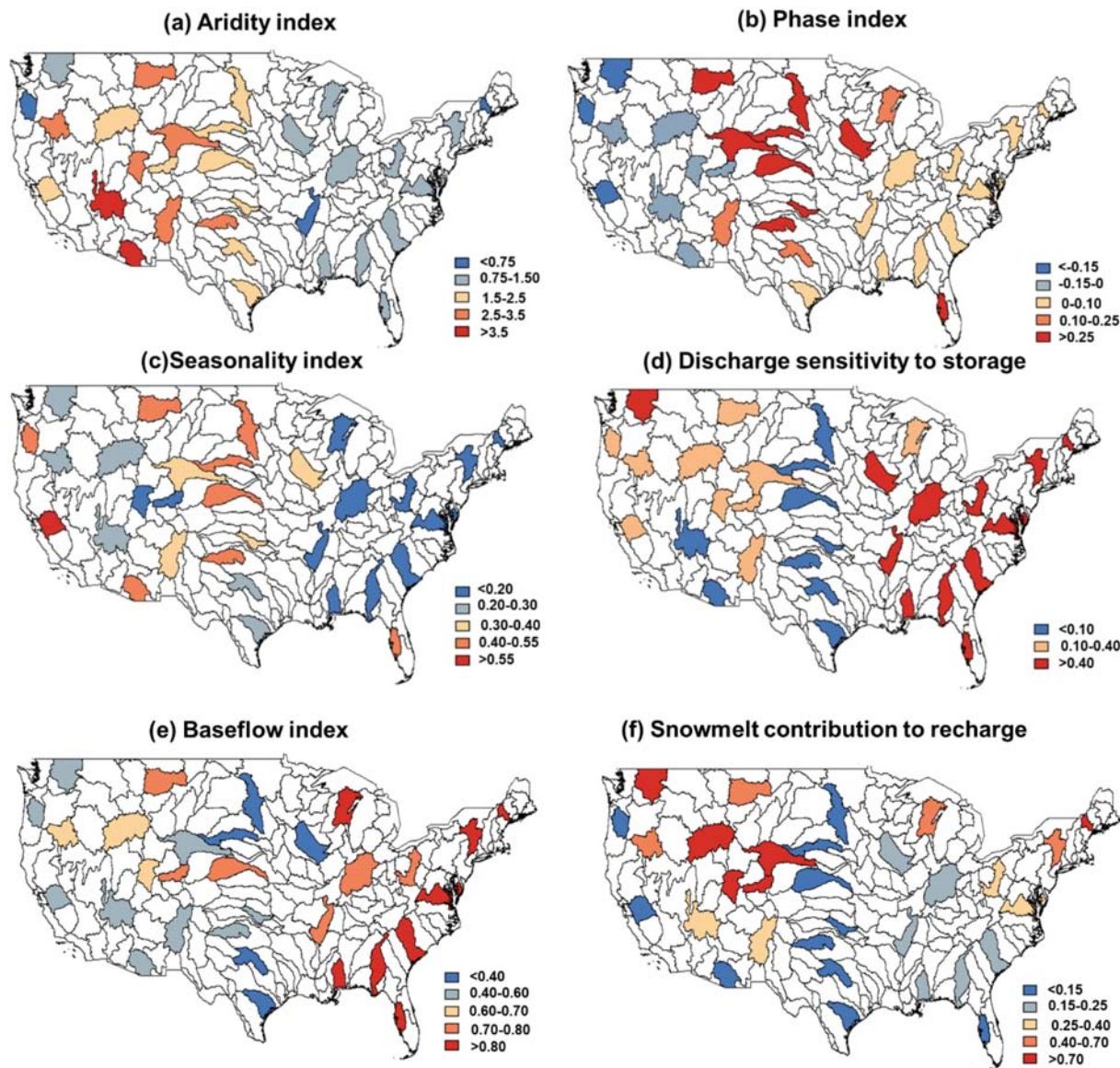


Figure 3. Climate and watershed properties of the watersheds considered in this study: (a) aridity index, (b) phase index, (c) seasonality index, (d) discharge sensitivity to storage, (e) baseflow index, and (f) partial correlation between snowmelt and recharge given precipitation.

spatial pattern in the values of propagation ratios can be observed in the figure, based on which we can classify the selected watersheds into three regions (Figure 1):

1. Western U.S. (WUS) region: comprising of WLM, UCR, OCB, SJQ, MIS, USN, GRE, COH, NPL, and RGE watersheds. Watersheds in this region show high intensity and have the longest duration of streamflow droughts relative to meteorological droughts among the selected watersheds.
2. Great Plains and southwest U.S. (GP-SW) region: comprising of REP, JAM, RED, LCM, LCO, GIL, CTX, NIB, and MBR watersheds. These watersheds have low intensity as well as duration of streamflow droughts relative to meteorological droughts.
3. Eastern U.S. (EUS) region: comprising of UMS, LMS, MIC, WBS, PAS, APC, PEC, EDS, UOH, LCH, UHD, and AND watersheds. These watersheds have streamflow droughts with high intensity but low duration relative to meteorological droughts.

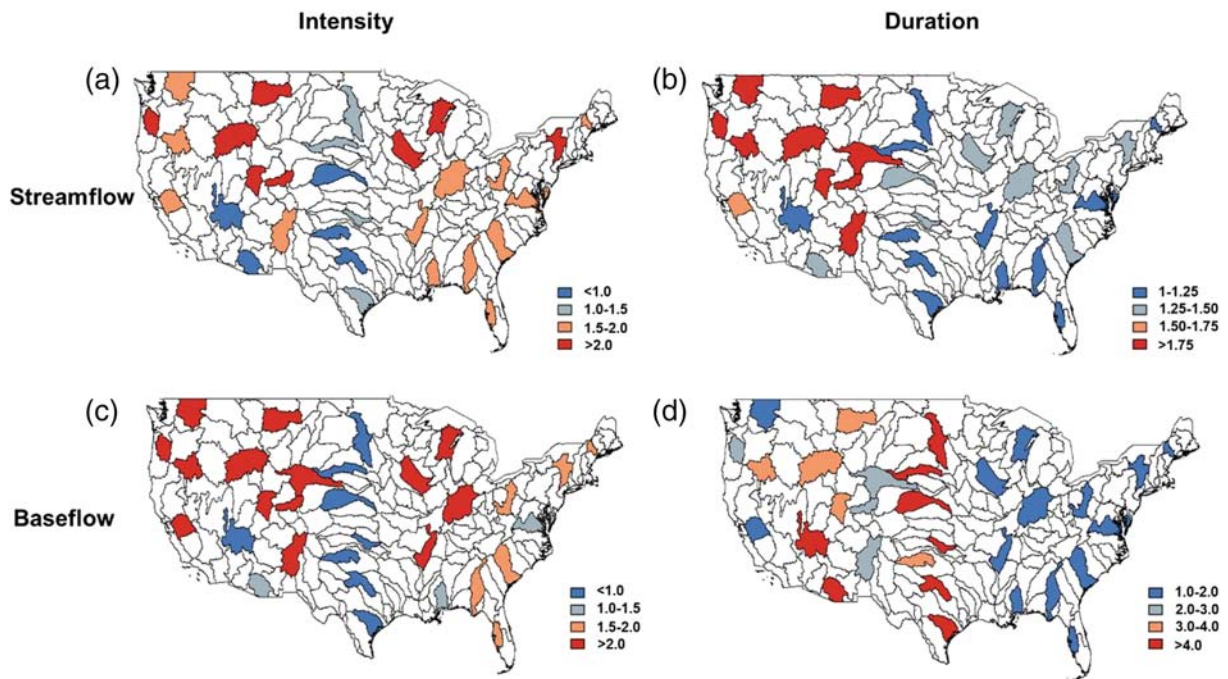


Figure 4. Propagation ratios (unitless) for (a) streamflow drought intensity, (b) streamflow drought duration, (c) baseflow drought intensity, and (d) baseflow drought duration.

An interesting observation to note from Figure 4 is that in the WUS and EUS watersheds, the characteristics of streamflow and baseflow droughts are very similar. However, for the watersheds in the GP-SW region, the characteristics of streamflow and baseflow droughts are quite different. In this region, baseflow droughts are very long compared to streamflow droughts (Figures 4b and 4d). It is also worth noting that watersheds in the three regions show similar hydrologic drought characteristics despite the variability in watershed properties such as soil type, land cover, and elevation (Figure S1), which shows a strong influence of climate properties on drought propagation.

While the results shown in Figure 4 are computed by using the 25th percentile threshold for defining droughts, we also repeat the analysis using the 20th and 30th percentile thresholds. The results of the sensitivity analysis are shown in Tables S2 and S3 of the supporting information, which show that there are no significant changes in the results compared to those with the 25th percentile.

3.2. Drought Propagation Mechanisms

Following the drought propagation ratio patterns discussed above, we analyze the drought propagation mechanisms using three watersheds, each of which is representative of the three regions identified in section 3.1. Through this analysis, we seek the understanding of the key differences in the physical mechanisms of drought propagation and how those differences contribute to different hydrologic drought characteristics in the three regions.

We select the COH watershed to describe the drought propagation mechanisms in the WUS region as this watershed has many features which are representative of the watersheds in this region, such as winter-dominated precipitation regime (Figure 3b), high discharge sensitivity to storage (Figure 3d), and high contribution of snowmelt to recharge generation (Figure 3f). The time series of different hydrologic processes and watershed storage for this watershed during a drought event in the observation period are shown in Figure 5. The first figure shows the climatology in terms of monthly average precipitation and monthly average potential ET (Figure 5a). As can be seen, the watershed has an out-of-phase-seasonality, i.e., more precipitation occurs in the winter season and summers are comparatively drier. The winter precipitation accumulates as snow, which melts in the spring season (Figures 5b and 5c) and generates recharge and

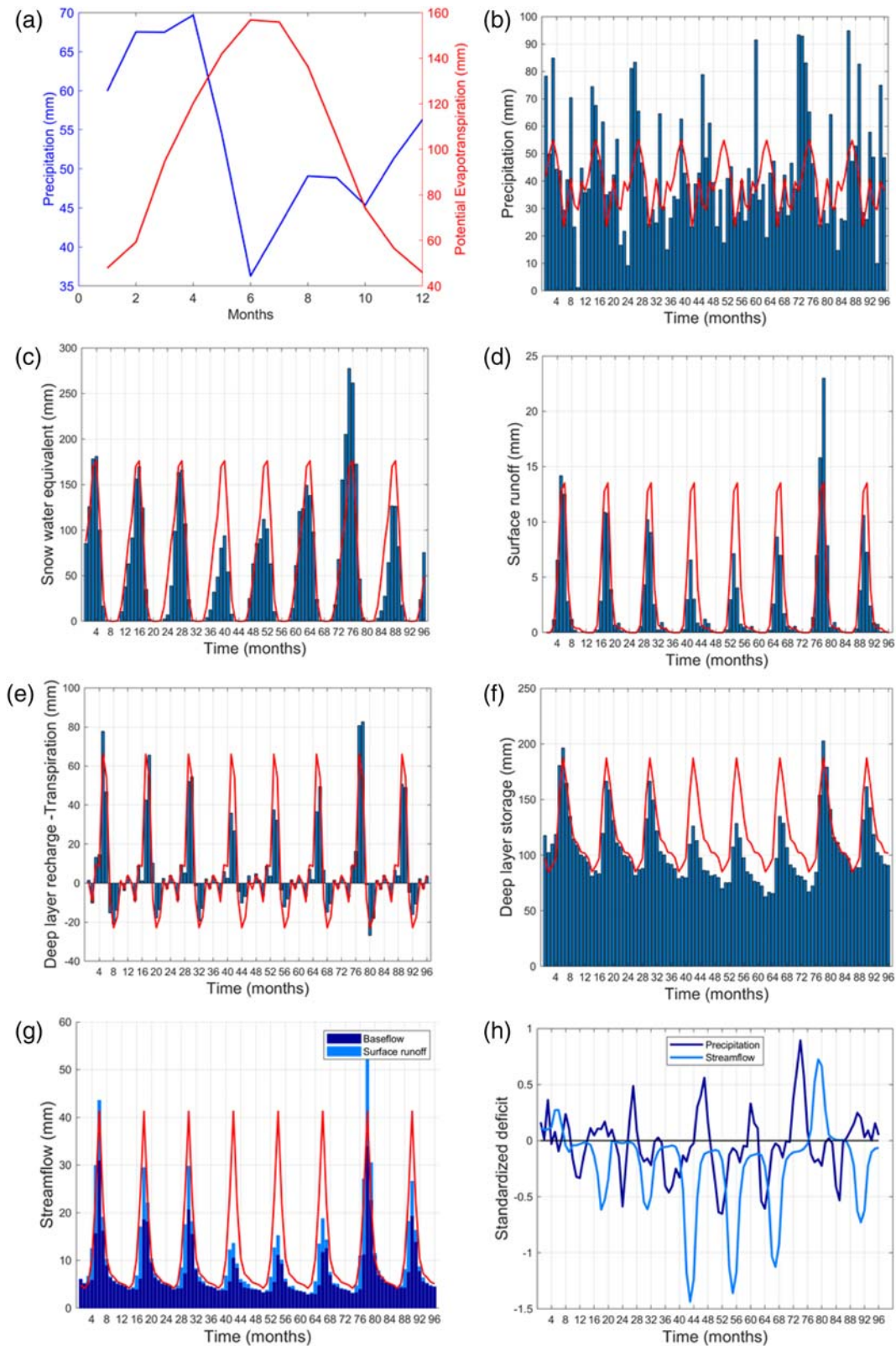


Figure 5. Drought propagation in the COH watershed: (a) monthly average precipitation (blue) and monthly average potential ET (red), (b) precipitation, (c) snow water equivalent, (d) surface runoff, (e) net recharge of deep layer storage calculated as the difference between recharge and transpiration, (f) deep layer storage, (g) streamflow, and (h) comparison of standardized time series of precipitation and streamflow. The red lines in panels (b)–(g) show the monthly threshold used for defining droughts for each variable.

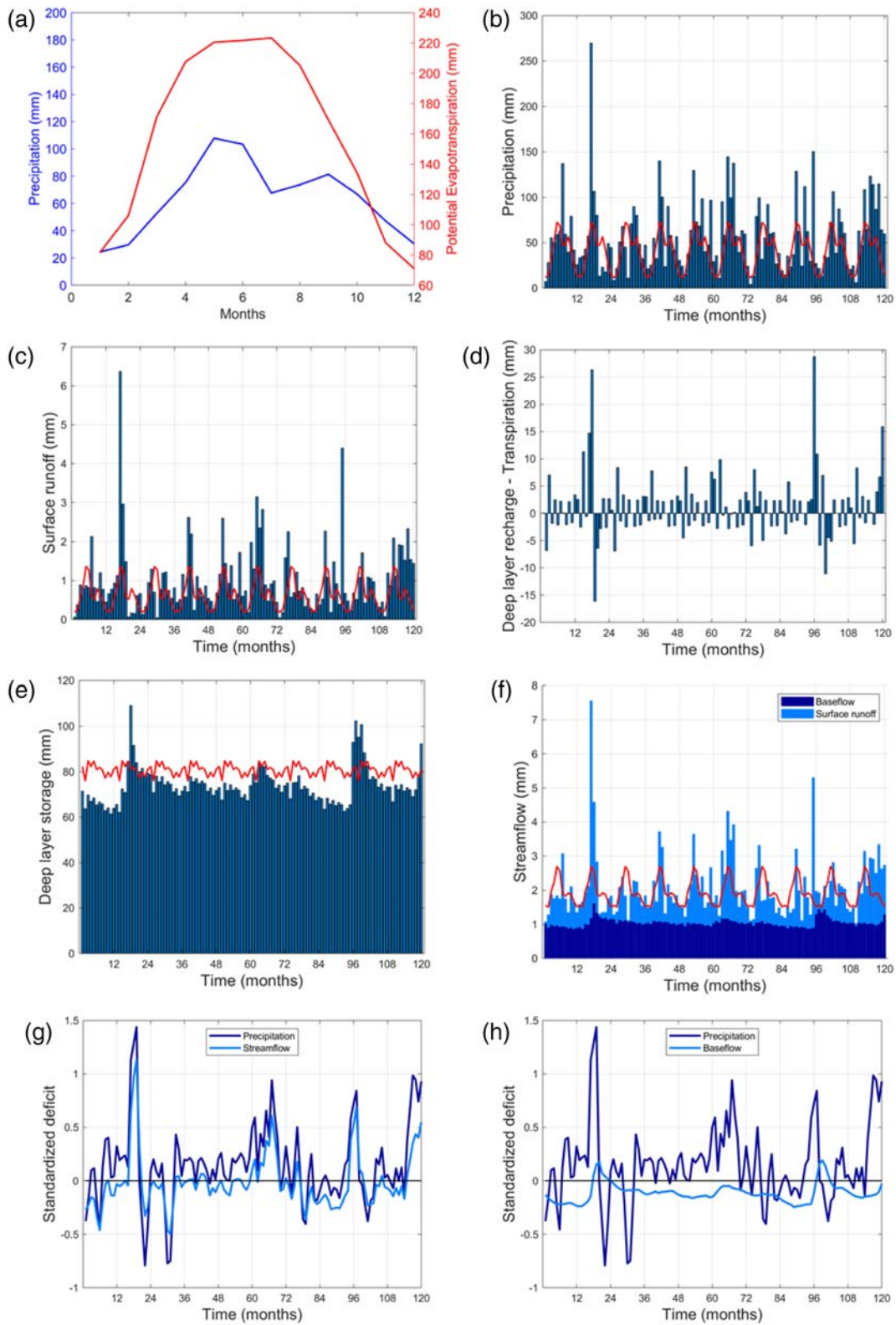


Figure 6. Drought propagation in the LCM watershed: (a) monthly average precipitation and monthly average potential ET, (b) precipitation, (c) surface runoff, (d) net recharge of deep layer calculated as the difference between recharge and transpiration, (e) deep layer storage, (f) streamflow, (g) comparison of standardized time series of precipitation and streamflow, and (h) comparison of standardized time series of precipitation and baseflow. The red lines in panels (b)–(f) show the monthly threshold used for defining droughts for each variable.

runoff in May and June (Figures 5d and 5e). The summer season has smaller precipitation and high potential ET, therefore most of the rainfall falling on the land-surface contributes to ET. Furthermore, the deep layer storage also loses water through transpiration in the summer season, as shown by the negative values of net recharge during the July–September period (Figure 5e). The strong seasonality in recharge contributes to strong seasonal variations in the deep layer storage, with a sharp increase in the storage in May and June and a subsequent decline in the remaining months of the year, during which no recharge occurs and the water in the storage is released through baseflow (Figure 5f). The baseflow responds to the strong seasonal changes in the deep layer storage, due to which baseflow also shows high seasonality, with high flows in spring and declining flows in the remaining part of the year. Therefore, the winter-dominated precipitation and snow lead to high seasonality in both surface runoff and baseflow and hence the streamflow (Figure 5g).

Since the recharge and runoff generation is restricted to a few months of the spring season, streamflow variability is very sensitive to the snow accumulation in the winter. Therefore, deficits in snowpack accumulation during droughts lead to significant seasonal deficits in streamflow (months 36–72 in Figure 5g), resulting in high intensity of streamflow droughts (Figure 5h). Furthermore, due to the strong seasonality in recharge, the watershed storage is replenished in just 2 months and it continues to fall in the rest of the year. As a result, when there is a shortage of spring recharge during droughts, the deficit in storage is carried forward from one year to another as the storage continues to decrease in the non-recharge season (Figure 5f). When such successive years of deficit build-up occurs, the recovery from hydrologic droughts is more difficult (Figure 5g), which leads to longer duration of streamflow droughts relative to meteorological droughts (Figure 5h).

Next, we select the LCM watershed to describe the drought propagation mechanisms in the GP-SW region considering its representative features among the watersheds in this region, such as arid climate (Figure 3a), summer-dominated precipitation regime (Figure 3b), low discharge sensitivity to storage (Figure 3d), and very little or no contribution of snowmelt to recharge generation (Figure 3f). The time series of different hydrologic processes and watershed storage for this watershed during a drought event in the observation period are shown in Figure 6. As can be seen from Figure 6a, in this watershed precipitation occurs predominantly in the summer season when potential ET is also high. Also, the magnitude of potential ET is much larger than the magnitude of precipitation (high aridity). Therefore, the climate can be characterized as an in-phase arid climate. Since precipitation occurs predominantly in the summer, most of the precipitation is lost to ET and the magnitudes of runoff and recharge generated in the watershed are relatively small. Although the magnitude of surface runoff is small, it is generated regularly, and its variability is strongly controlled by precipitation (Figures 6b and 6c). However, unlike runoff, recharge is generated only in the months with very high rainfall (Figure 6d). Due to the high amount of water lost to ET, the soil moisture levels are quite low and very little recharge can be generated due to the nonlinear relationship between soil moisture and recharge (Brooks & Corey, 1964). As a result, recharge can be generated only in years with very high rainfall in the summer months, e.g., months 16–19, or months with some winter rainfall such as month 96 (Figures 6b and 6d). The infrequent recharge events due to low soil moisture levels lead to very long periods of low baseflows, which explains the long duration of baseflow droughts relative to meteorological droughts in this watershed (Figure 6h). Although the baseflow droughts are long, the variability in baseflow in response to the changes in deep layer storage is small (Figures 6e and 6f), which explains the low intensity of baseflow droughts (Figure 6h). For instance, the recharge generated in months 16–19 and in month 96 increases the deep layer storage (Figure 6e) but there is a very small change in baseflow (Figure 6f) and most of the recharged water is lost to transpiration as shown by the negative values of net recharge following the recharge events in Figure 6d. Due to the low variability in baseflow, the surface runoff becomes the dominant contributor to streamflow variability. Since the variability in surface runoff is controlled strongly by precipitation, the streamflow and meteorological droughts have similar duration in this watershed (Figure 6g). Furthermore, the nearly constant contribution of baseflow to streamflow (Figure 6h) acts as a buffer during drought events, which results in low intensity of streamflow droughts (Figure 6g) compared to meteorological droughts.

Finally, we take the APC watershed as an example to describe the drought propagation mechanisms in the EUS region because of its many features which characterize the watersheds in this region, such as humid climate (Figure 3a), low seasonality of precipitation (Figure 3c), high discharge sensitivity to storage

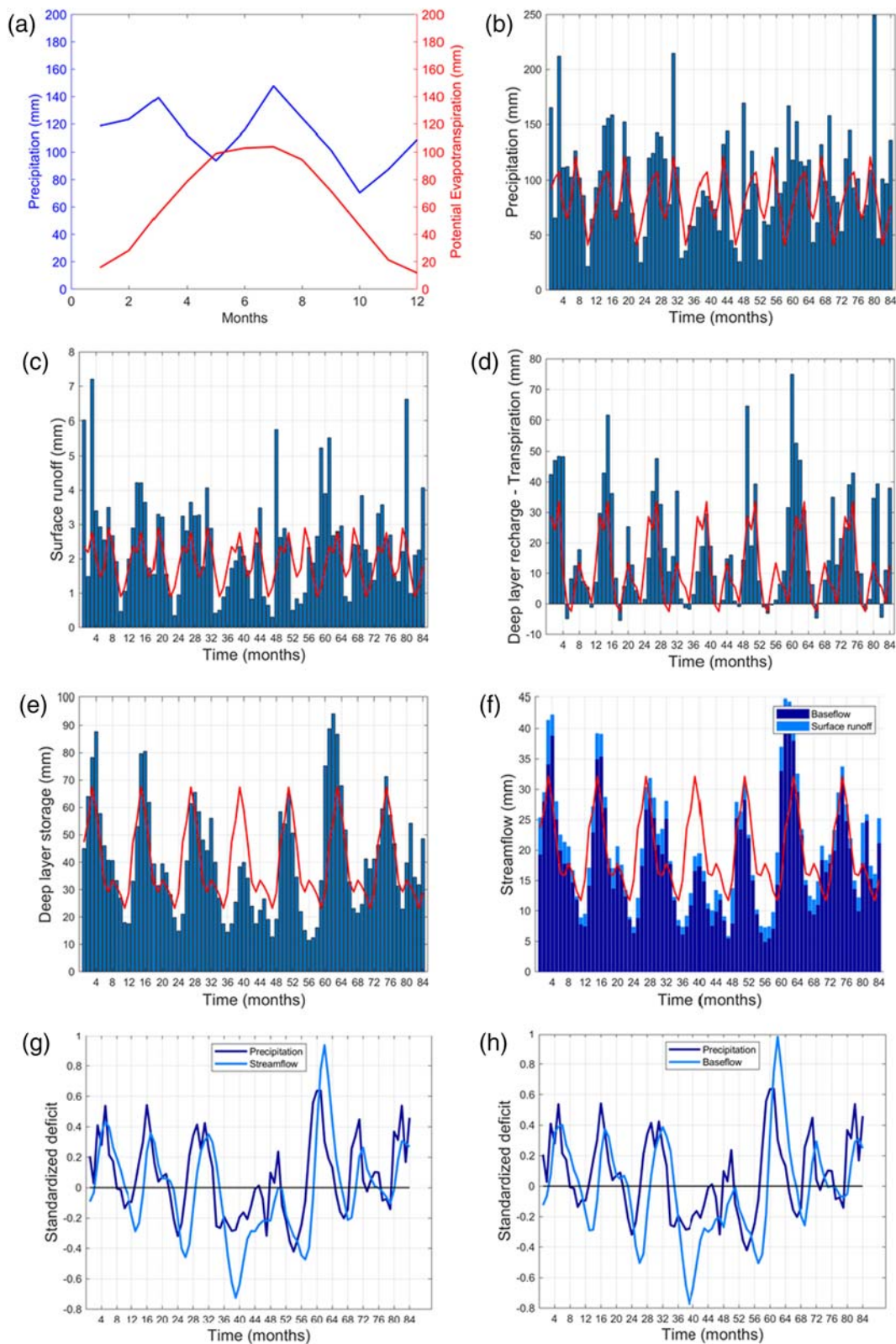


Figure 7. Drought propagation in APC watershed: (a) monthly average precipitation and monthly average potential ET, (b) precipitation, (c) surface runoff, (d) net recharge of deep layer storage calculated as the difference between recharge and transpiration, (e) deep layer storage, (f) streamflow, (g) comparison of standardized time series of precipitation and streamflow, and (h) comparison of standardized time series of precipitation and baseflow. The red lines in panels (b)–(f) show the monthly threshold used for defining droughts for each variable.

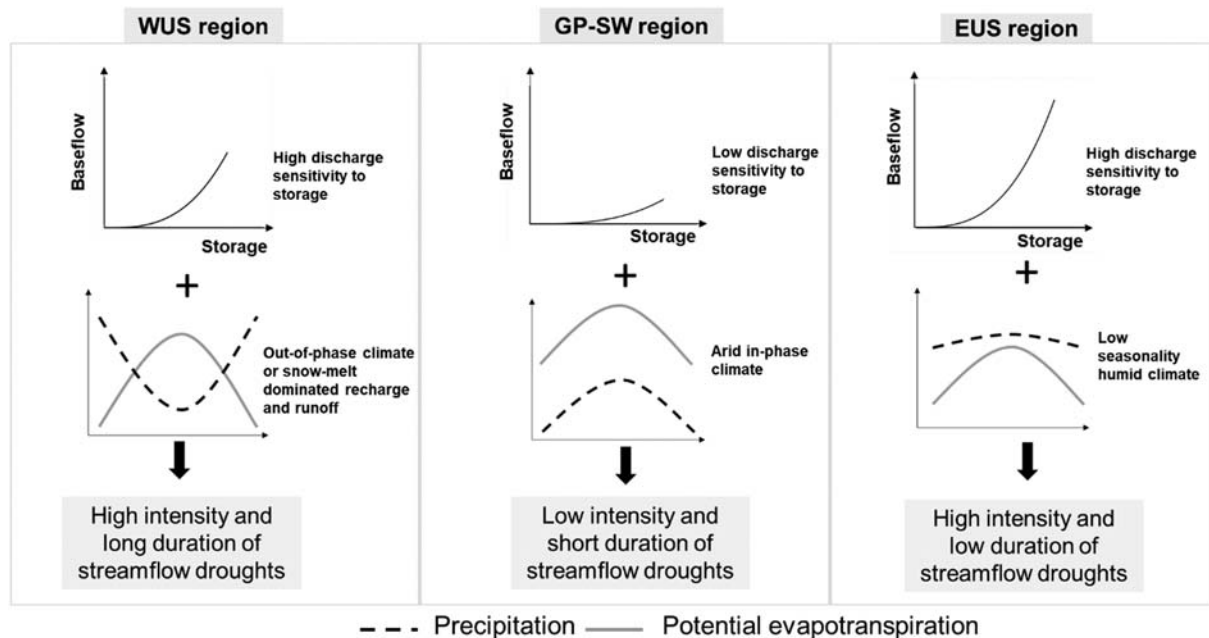


Figure 8. Dominant controls for drought propagation for watersheds in the WUS, the GP-SW, and the EUS regions.

(Figure 3d), and low contribution of snowmelt to recharge generation (Figure 3f). The time series of different hydrologic processes and watershed storage for this watershed during a drought event in the observation period are shown in Figure 7. The watershed receives precipitation throughout the year and the magnitude of precipitation is greater than that of potential ET (Figures 7a and 7b). Due to the low seasonality and the low aridity of the climate, the precipitation occurring throughout the year can keep the soil wet, and recharge and runoff can be generated in almost all the months of the year (Figures 7c and 7d). The strong control of precipitation over recharge subsequently leads to strong variations in the deep layer storage in response to the precipitation variability (Figure 7e). The baseflow responds actively to the variations in the storage (Figure 7f), which ultimately results in high intensity of baseflow droughts (Figure 7h). Due to the strong influence of precipitation over both surface runoff and baseflow, the onset and recovery of streamflow droughts is controlled by precipitation, resulting in a similar duration of meteorological and streamflow droughts (Figure 7g).

3.3. Factors Affecting Drought Propagation

3.3.1. Watersheds of the GP-SW Region

The watersheds in the Great Plains region have arid climates (Figure 3a) with summer-dominated precipitation (Figure 3b). Due to these conditions, most of the soil moisture is lost to ET and recharge is generated only during events with very high precipitation. Furthermore, these watersheds also show very low sensitivity of discharge relative to the watershed storage (Figure 3d). Therefore, in this region both the climate and watershed properties contribute to significant dampening of climate variability to generate a smooth baseflow (Figure 6). On one hand, the low variability in baseflow reduces the intensity of streamflow droughts; on the other hand, it implies that the streamflow variability is dominated by surface runoff which results in faster recovery and hence short duration of streamflow droughts. Thus, the low sensitivity of discharge to storage, combined with the arid in-phase climate in this region, results in low duration and intensity of streamflow droughts relative to meteorological droughts (Figure 8). The low sensitivity of discharge to storage is also the main reason for different characteristics of streamflow and baseflow droughts in this region (Figure 4).

The watersheds in southwest United States (LCO and GIL) receive more precipitation in the winter (Figure 3b); however, the highly arid climates of these watersheds result in large ET losses and limited recharge generation (Figure S2), similar to the Great Plains watersheds. Furthermore, these watersheds

also have very low sensitivity of discharge to storage which contributes to low intensity and short duration of streamflow droughts in the region (Figure S2).

3.3.2. Watersheds in the WUS Region

These watersheds, unlike those in the GP-SW region, have higher sensitivity of discharge relative to the watershed storage (Figure 3d), which means that in these watersheds, the streamflow varies more strongly in response to the variations in the watershed storage. Therefore, in these watersheds, the climate variability is propagated to streamflow variability, resulting in high intensity of streamflow droughts (Figure 4). In the WUS region, most of the watersheds have winter-dominated precipitation as shown by the negative values of the phase index in Figure 3b, and a significant portion of the winter precipitation falls as snow. The snowmelt in the spring season leads to strong seasonal variations in recharge, runoff, and watershed storage. The watershed storage increases sharply due to snowmelt in the spring and decreases in the remaining months of the year when little or no recharge is generated. Therefore, in drought years, the shortage in watershed storage caused by lack of recharge is carried forward from one year to another due to the continuous loss of storage in the non-recharge season. Due to the high sensitivity of discharge to storage, the decline in watershed storage in successive years contributes to decline in streamflow, making the recovery of hydrologic droughts more and more difficult (Figure 5). Thus, the combination of high discharge sensitivity to storage and climate with winter-dominated precipitation leads to streamflow droughts which are more intense and longer than meteorological droughts in the watersheds of the WUS region (Figure 8). However, there are some watersheds in this region, such as the SJQ and WLM watersheds, in which snowmelt does not contribute significantly to recharge generation (Figure 3f), even though these watersheds also have winter-dominated precipitation regimes (Figure 3b). The winter rainfall in these watersheds also results in highly seasonal generation of recharge and runoff, which contributes to long duration and high intensity of hydrologic droughts as shown in Figure S3 for the SJQ watershed.

It is interesting to note that MIS, NPL, and RGE watersheds have positive phase index (Figure 3b) and high aridity index (Figure 3a), which is more similar to the climate of watersheds in the GP-SW region. However, these two watersheds still show hydrologic drought characteristics similar to those in the WUS region (Figure 4). This is because these watersheds receive a significant fraction of the precipitation as snow, which plays an important role in the recharge and runoff generation (Figure 3f). Therefore, despite the summer-dominated precipitation, recharge and runoff are generated predominantly in the spring season due to snowmelt. The snowmelt-driven recharge and runoff cause similar seasonal variations in streamflow and watershed storage during droughts, as those observed in the watersheds of the WUS region with snowmelt-driven recharge, as shown in Figure S4 for the RGE watershed.

3.3.3. Watersheds in the EUS Region

These watersheds have the strongest sensitivity of discharge to storage among the selected watersheds (Figure 3d), which results in strong variability of streamflow in response to the climate variability, and therefore in high intensity of streamflow droughts. However, the watersheds in this region have very weak seasonality in precipitation (Figure 3c) and humid climate (Figure 3a), due to which both recharge and runoff can be generated in almost all the months of a year. Therefore, the onset and recovery of streamflow droughts are both strongly controlled by meteorological droughts, resulting in similar duration of meteorological and streamflow droughts. Hence, due to the combination of low-seasonality humid climate and high discharge sensitivity to storage, the streamflow droughts in this region have higher intensity and similar duration as meteorological droughts (Figure 8).

An exception to the climate characteristics of most EUS watersheds is the PEC watershed, which has summer-dominated precipitation regime (Figure 3b) and also higher seasonality than most EUS watersheds (Figure 3c). However, due to the humid climate, precipitation has a strong control over the onset and recovery of hydrologic droughts and therefore meteorological and hydrological droughts have similar duration like other watersheds of the EUS region (Figure S5).

It is important to note that similar to the watersheds of the WUS region, in some watersheds in the EUS region such as AND, MIC, and UHD, snowmelt plays an important role in recharge generation (Figure 3f), but these watersheds do not show characteristics of hydrologic droughts like the WUS watersheds. The reason for this difference is that in the WUS watersheds, recharge and runoff generation is highly seasonal and can take place only in the winter or spring due to the strong seasonality in precipitation and

higher aridity, which leads to long streamflow droughts in this region. However, in the EUS watersheds, recharge and runoff generation is not confined to the snowmelt season and can also take place in other months of a year due to the low seasonality in precipitation and humid climate, which results in shorter streamflow droughts in this region, as shown in Figure S6 for the MIC watershed.

4. Discussion

4.1. Hydrologic Drought Characteristics

The propagation mechanisms identified in the three regions of the CONUS in this study correspond to the three drought propagation mechanisms identified in Apurv et al. (2017). In the WUS region, we find hydrologic droughts which are more intense and longer than meteorological droughts. The hydrologic droughts in the watersheds of this region are characterized by large seasonal deficits in streamflow, which can accumulate in successive years due to the strong seasonal cycles in recharge and runoff generation. This propagation mechanism was called as “Type 1” pattern in the study by Apurv et al. (2017), and it was proposed that this pattern can arise in watersheds with winter-dominated precipitation, which is consistent with the finding of this study. In the Great Plains region, we find that the arid climate and summer-dominated precipitation regime lead to very low soil moisture levels, due to which recharge can be generated only during high rainfall events, resulting in very long baseflow droughts. This pattern was defined in Apurv et al. (2017) as “Type 2” propagation mechanism, arising in arid and in-phase climates. This pattern has also been identified by Van Loon and Van Lanen (2012) as the “wet to dry season droughts”. Finally, in the watersheds of the EUS region, we find that the hydrologic droughts have similar duration to meteorological droughts due to the strong control of precipitation over runoff and recharge generation. This pattern was defined as “Type 3” pattern arising in low seasonality humid climates in Apurv et al. (2017). This pattern was also identified by Van Loon and Van Lanen (2012) and was referred to as “classical rainfall deficit drought”.

However, we also find some important differences in the drought propagation mechanisms from those identified in previous studies. First, Apurv et al. (2017) showed that the “Type 1” propagation mechanism can arise in climates with winter-dominated precipitation. In the present study, we find that this pattern can also arise in summer-dominated precipitation regimes if the watershed receives some snowfall in the winter and the recharge and runoff cycles are driven by the snowmelt, as found in the case of MIS, NPL, and RGE watersheds. Thus, along with the timing of rainfall, snow is also an important factor for “Type 1” drought propagation mechanism. Second, the study by Apurv et al. (2017) focused on the characteristics of baseflow droughts under different climatic conditions; while in the current study, we analyze the characteristics of both baseflow and streamflow droughts to understand the role of both surface and subsurface flows in drought propagation. Here, we show that in watersheds with high sensitivity of discharge to storage, baseflow and streamflow droughts have similar characteristics; however, for watersheds with low sensitivity of discharge to storage, the characteristics of streamflow and baseflow droughts can be quite different, as found in the cases of watersheds in the GP-SW region. Finally, Apurv et al. (2017) and Van Loon and Van Lanen (2012), which both addressed process-based classification of drought propagation mechanisms, classified hydrologic droughts based on climate properties only. In this study, we show that the discharge sensitivity to storage is an important watershed property which needs to be considered for process-based classification of drought propagation mechanisms together with climate properties, as further discussed in section 4.2.

It should be noted that a recent study by Konapala and Mishra (2020) also analyzed the spatial patterns of streamflow drought characteristics in the United States. They found shorter duration and higher intensity of streamflow droughts in eastern parts of the United States, including Midwest, Southeast United States, and Northeast United States and longer duration of streamflow droughts in Western United States, which is consistent with the findings of our study. However, they also identified short hydrologic droughts in the Pacific Northwest and longer hydrologic droughts in some of the watersheds in Southeast United States, which is opposite of what we find here. The difference may be due to the difference in the methodology adopted in the two studies. In our study, we have studied the spatial patterns of propagation ratios, which measure the duration and intensity of streamflow droughts relative to those of meteorological droughts;

whereas Konapala and Mishra (2020) used the average duration and intensity of streamflow droughts for studying the spatial patterns of hydrologic droughts.

4.2. The Role of Climate and Watershed Properties

Several previous studies have investigated the factors which control drought propagation. Watershed elevation has been identified as an important factor for drought propagation in these studies. For instance, Konapala and Mishra (2020) found that watersheds with higher elevation have longer duration of hydrologic droughts in the United States. Peña-Gallardo et al. (2019) found longer lag times between recovery of meteorological and hydrologic droughts in watersheds at higher elevations in the United States. Van Loon and Laaha (2015) found that watersheds at higher elevation experience more deficits in streamflow during hydrologic droughts. The results of our study suggest that this could be due to the presence of snow in high elevation watersheds. Most of the river basins in the WUS have high elevations (Figure S1), and the low temperatures at higher elevations result in the accumulation of the winter precipitation as snow. As a result, snowmelt plays an important role in highly seasonal recharge generation (Figure 3f). As demonstrated for the case of the COH watershed (Figure 5), the high seasonality in recharge and runoff generation can lead to streamflow droughts which are more intense and longer than meteorological droughts.

The BFI has also been identified as an important factor which influences drought propagation in previous studies (Barker et al., 2016; Konapala & Mishra, 2020; Van Loon & Laaha, 2015). These studies have found that watersheds with larger BFI have longer streamflow droughts due to the longer duration of baseflow droughts. Indeed, we find that baseflow contributes to longer duration of streamflow droughts in the watersheds of the WUS region due to the accumulation of storage deficits during droughts (Figure 5). In the GP-SW region too, we find that baseflow droughts have very long durations (Figure 4d); however, the duration of streamflow droughts is much shorter (Figure 4b) due to the low variability of baseflow and dominant contribution of surface runoff in the hydrologic drought recovery (Figure 6). Thus, baseflow does not contribute to longer duration of streamflow droughts in this region due to the low sensitivity of discharge to storage. In the EUS watersheds, the duration of baseflow droughts is short due to the strong control of precipitation over recharge generation and the strong sensitivity of discharge to storage, which allows baseflow to recover quickly from hydrologic droughts. As a result, watersheds in this region do not have long streamflow droughts despite the high values of VIC-simulated BFI (Figure 3e) in these watersheds. Therefore, referring to the findings of previous studies, here we show that high BFI values may not necessarily lead to longer streamflow droughts and the influence of BFI on the streamflow drought duration depends on the climate properties and the storage-discharge relationship in the watershed.

In this study, we identify the storage-discharge relationship to be a key hydrologic property of watersheds that influences drought propagation. Several previous studies analyzed the storage-discharge relationships in the watersheds of the United States. Beck et al. (2013) studied the streamflow recession characteristics of 3,394 watersheds from around the world using the observed streamflow data, which included many watersheds from the United States. They found that the watersheds in the Great Plains region and southwest United States had shorter recession times whereas watersheds in western and eastern United States had longer recession times. Thus, in the Great Plains and southwest U.S. watersheds, streamflow recedes rapidly following high flow events due to the nearly constant contribution of baseflow during and after storm events; whereas in watersheds of western and eastern United States, streamflow recedes more gradually due to the variability of baseflow. A similar pattern was also found in a recent study by Tashie, Pavelsky, and Band (2020), who analyzed the streamflow recession behavior of 1,027 streams across the CONUS. Zhang et al. (2019) reported slower response of groundwater storage in response to precipitation in the Great Plains region and faster response in the western and eastern United States. The patterns identified in these previous studies are consistent with the spatial pattern of storage-discharge relationships obtained from the VIC model outputs, as shown Figure 3f.

Previous studies have found a significant influence of climate properties on the storage-discharge relationships of watersheds. In particular, previous studies have found weak sensitivity of discharge to storage in regions with high aridity index and climates with summer dominated precipitation. Beck et al. (2013) explained that under these climatic conditions, most of the watershed storage after recharge events was lost to ET due to the high atmospheric demand of water; as a result, baseflow did not vary significantly in response to the storage changes. Recently, Tashie, Pavelsky, and Emanuel (2020) also demonstrated the

important role played by ET in affecting the baseflow recession characteristics in the watershed of the United States. Many other studies have also found climate properties such as aridity index, mean annual precipitation, and timing of precipitation to influence the streamflow recession behavior in watersheds (Cooper et al., 2018; Krakauer & Temimi, 2011; Peña-Arancibia et al., 2010; van Dijk, 2010).

Besides climate properties, watershed properties such as topography and geology can also influence the storage-discharge relationship (Bloomfield et al., 2009; Kroll et al., 2004; Laaha & Blöschl, 2006). Watersheds with steeper slopes have been found to have a higher sensitivity of discharge to storage. For instance, Sproles et al. (2015) showed that the Upper Columbia watershed in the northwest United States, which has a steeper topography, exhibited a stronger variation in baseflow in response to the watershed storage as compared to the Snake river basin which has a flatter terrain. Tashie, Pavelsky, and Band (2020) showed that the flat terrain in the Great Plains region also contributed to lower variability of baseflow in the watersheds of this region. In terms of geological properties, Stoelzle et al. (2014) showed that in watersheds with karstic and fractured aquifers, baseflow shows short-term response to variations in storage whereas the watersheds with porous aquifers show a more delayed response of baseflow. There is significant variation in the geological properties of aquifers in different parts of the CONUS (Miller, 1999). The Upper Columbia and Upper Snake watersheds in the Northwest United States are underlain by basaltic rock aquifers which have high permeability; whereas the aquifer in the Upper Hudson watershed consists of fractured crystalline rocks. San Joaquin watershed is underlain by unconsolidated marine sediment deposits of sand and gravels whereas Willamette, Wabash, and Lake Michigan watersheds overlie aquifers composed of unconsolidated glacial deposits. In the southeast United States, Peace, Apalachicola, and Edisto watersheds are underlain by Floridan and surficial aquifer systems, respectively, which contain layers of limestone, leading to strong hydraulic connection between groundwater and streamflow in this region (Couch et al., 1996). Hence, a strong storage-discharge relationship can be found in the watersheds in this region. Rivers in the Great Plains region such as Republican, Cimarron, and Brazos rivers overlie the High Plains aquifer, an alluvial aquifer consisting of unconsolidated sand and gravels. Watersheds in the arid southwest United States such as the Gila river is also underlain by alluvial aquifers. The slow response of porous aquifers, as demonstrated by Stoelzle et al. (2014) and Haslinger et al. (2014), could also be a contributing factor for the lower variability of baseflow in the Great Plains and southwest U.S. watersheds besides the high aridity in climate of these regions. Thus, the storage-discharge relationship represents a composite impact of climatic, geologic, and topographic characteristics of watersheds.

Furthermore, it is important to note that the spatial scale of analysis plays an important role in the identification of the dominant controlling factors for drought propagation. In this study, we have analyzed drought propagation at the national scale by selecting watersheds from different regions of the CONUS and the spatial pattern of the characteristics of hydrologic droughts are found to coincide with the pattern of climate properties. However, many of the previous studies have studied drought propagation controlling factors for multiple watersheds or subwatersheds lying within a single region. For example, Veetil and Mishra (2020) found the stream order as an important variable controlling drought propagation in the sub watersheds of the Savannah river basin in the southeast United States. Similarly, Yang et al. (2017) identified valley bottom flatness as a key variable controlling hydrologic drought recovery in watersheds of southeastern Australia. In general, climate properties have a more significant influence on the patterns of drought propagation at larger scales and watershed properties play a more significant role for watersheds lying within the same region.

4.3. Limitations

An important limitation of the analysis presented in this study is that it is based on the outputs of a hydrologic model, which are subject to uncertainties that may arise from the model structure, parameters, and input data (Refsgaard & Storm, 1990). Furthermore, as shown in a recent study by Melsen and Guse (2019), the most influential hydrologic model parameters for hydrologic drought simulation can vary significantly from one model to another due to the model structure differences. Therefore, the results of a modeling-based study like those presented in this study should be treated as a source of hypotheses, and validation with observed data is required.

The VIC model uses a combination of calibrated and physically based parameters. The physically based parameters include soil characteristics such as saturated hydraulic conductivity, field capacity, porosity, and land cover characteristics such as leaf area index (LAI), rooting depth, architectural, and stomal resistance, which are specified using observation-derived data sets. As described in section 2.1, these parameters have been derived from multiple sources, which can be a source of uncertainty in the model outputs. The calibrated parameters include soil parameters describing depth of soil layers, soil infiltration capacity, and baseflow drainage. In the Livneh et al. (2013) data set, these parameters were adjusted to match the simulated runoff with the observed runoff at the outlets of 12 large river basins in the CONUS, which resulted in large regions with similar model parameters and strong discontinuities in the parameter fields at the basin boundaries (Mizukami et al., 2017).

However, several studies which have evaluated the water balance simulated by the VIC model for the CONUS have shown that the model is able to capture most features of the hydrologic cycle reasonably well. For example, Xia et al. (2012) evaluated the characteristics of the VIC-simulated streamflow for 961 small watersheds and eight major watersheds in the CONUS. They found high correlation coefficients between modeled and observed monthly streamflow in most regions of the CONUS, thereby demonstrating that the model is able to capture the seasonal dynamics of water balance in most parts of the country. They showed that the VIC model captures the seasonal variation of streamflow particularly well in the northwest and northeast United States where snowmelt is an important contributor to streamflow, and thus the VIC model is able to simulate snow related processes with reasonable accuracy. However, several studies have shown that the performance of the VIC model in simulating streamflow in the arid regions of central United States is poorer as compared to the humid regions (Mizukami et al., 2017; Oubeidillah et al., 2014; Xia et al., 2012), which is due to several contributing factors such as errors in forcing data, anthropogenic effects on streamflow in the region and limitations of VIC model in modeling the physics of ET-related processes, which play an important role in runoff generation in arid regions (Oubeidillah et al., 2014; Troy et al., 2008; Xia et al., 2012).

The study by Cai et al. (2014) compared the storage variability simulated by the VIC model with the observed total terrestrial storage variability estimated using the Gravity Recovery and Climate Experiment (GRACE) data set. They showed that the VIC model captures the spatial pattern of storage variability in the CONUS quite well with high storage variability in the northwestern and southeastern United States and low variability in central and southwestern United States. Furthermore, we also find that the spatial pattern of discharge sensitivity to storage (Figure 3d) is consistent with previous studies which have analyzed the storage-baseflow relationship based on streamflow recession analysis (Beck et al., 2013; Tashie, Pavelsky, & Band, 2020).

The study by Yang et al. (2019) showed that VIC simulation with LDAS parameters underestimates the BFI in northwest United States and overestimates it in southeast United States. In the northwest United States, we find that baseflow contributes to longer duration of streamflow droughts due to the accumulation of storage deficits during the drought events (Figure 5). Therefore, the underestimation of BFI by VIC model implies that the duration of streamflow droughts could be longer than what we find in this study. However, in the southeast United States both recharge and runoff generation are strongly controlled by precipitation (Figure 7); therefore, the overestimation of BFI by VIC model in this region is not likely to affect the characteristics of hydrologic droughts identified in this region in our study.

Another limitation of our study is that while we focus on understanding the role of climate and watershed properties on drought propagation, vegetation can also play an important role in the drought propagation process (Bloomfield et al., 2019). Vegetation can not only transpire water from the soil layer but also from groundwater, thereby affecting the partitioning of groundwater storage between transpiration and baseflow (Maxwell & Condon, 2016). Furthermore, different vegetation covers can have very different water utilization strategies during droughts. For example, Koirala et al. (2017) showed that while the productivity of shrubs decreased during drought periods with deeper groundwater tables, trees were able to maintain high productivity levels during droughts by using their deep roots to draw water for transpiration. Also, the response of vegetation to droughts can be very different in different climate conditions (Dong et al., 2019). With a relatively simple representation of vegetation

included in the VIC model, such complex water use behavior of vegetation may not be captured by the model. Hence, the role of vegetation on drought propagation should be studied in more detail in future studies.

5. Conclusions

In this study, we analyze the outputs of a hydrologic model to gain process-based understanding of the drought propagation mechanisms of a number of watersheds selected from different regions of the CONUS. We first analyze the spatial patterns in the characteristics of hydrologic droughts in the selected watersheds and identify three regions in the CONUS with different characteristics of hydrologic droughts: (1) western United States, where watersheds have both high intensity and duration of streamflow droughts relative to meteorological droughts; (2) Great Plains and southwest United States, where watersheds have intensity lower than meteorological droughts but similar duration as meteorological droughts; (3) eastern United States, where watersheds have higher intensity than meteorological droughts but similar duration as meteorological droughts. Following that, major controlling factors which contributed to these patterns are analyzed by comparing the hydrologic processes in the watersheds of these three regions. We find that the spatial pattern of hydrologic drought characteristics is strongly influenced by climate properties such as aridity, seasonality, and timing of precipitation. Furthermore, we identify the storage-discharge relationship as a key hydrologic property of the watersheds which controls the intensity of hydrologic droughts. The combination of high discharge sensitivity to storage and climate with winter-dominated precipitation leads to intense and long streamflow droughts relative to meteorological droughts in the watersheds of the western United States, while the combination of low-seasonality humid climate and high discharge sensitivity to storage leads to streamflow droughts with high intensity and similar duration as meteorological droughts in the watersheds of the eastern United States. In the watersheds of the Great Plains and southwest United States, the low discharge sensitivity to storage combined with the arid climate results in low duration and intensity of streamflow droughts. The low sensitivity of discharge to storage is also the main reason for different characteristics of streamflow and baseflow droughts in this region.

The differences in the characteristics of hydrologic droughts identified in this study show that the impacts of droughts on water resources can be very different in different regions of the CONUS in terms of the magnitude and duration of deficits in water availability. Therefore, the results of this study can help water resource managers in designing more effective drought management strategies by taking the physical mechanisms into consideration. The conceptual understanding of drought propagation mechanisms and drought management implications in the CONUS presented in this study can be applied to other parts of the world for better characterization of hydrologic droughts in regions with different climatic and hydrologic characteristics. Furthermore, the classification of drought propagation mechanisms based on climate and watershed properties can facilitate the comparison and synthesis of drought propagation studies in different parts of the world.

Data Availability Statement

The data for watershed properties were accessed from the NLDAS data set: soil properties were taken online (<https://ldas.gsfc.nasa.gov/nldas/soils>), land cover data were taken online (<https://ldas.gsfc.nasa.gov/index.php/nldas/vegetation-class>), and elevation data were taken online (<https://ldas.gsfc.nasa.gov/nldas/elevation>). The analysis performed in this study used the Livneh et al. (2013) data set, which was accessed through the following websites, the precipitation data were taken online (<https://psl.noaa.gov/data/gridded/data.livneh.metvars.html>) and the VIC model outputs were taken online (<https://psl.noaa.gov/data/gridded/data.livneh.fluxvars.html>).

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